EXPRIMENT-15

15.) Letter Frequency Attack on Additive Cipher

Aim:

To write a Python program to perform a letter frequency attack on an additive cipher.

PROGRAM:

```
def frequency_attack(cipher):

    cipher = cipher.lower()

    for shift in range(26):
        result = ""

        for ch in cipher:
            if ch.isalpha():
                value = ord(ch) - 97
                value = (value - shift) % 26
                result += chr(value + 97)
            else:
                result += ch

        print("Shift", shift, ":", result)


message = input("Enter Cipher Text: ")

frequency_attack(message)
```

OUTPUT:

```
Output
Shift 0 : khoor
Shift 1 : jgnnq
Shift 2 : ifmmp
Shift 3 : hello
Shift 4 : gdkkn
Shift 5 : fcjjm
Shift 6 : ebiil
Shift 7 : dahhk
Shift 8 : czggj
Shift 9 : byffi
Shift 10 : axeeh
Shift 11 : zwddg
Shift 12 : yvccf
Shift 13 : xubbe
Shift 14 : wtaad
Shift 15 : vszzc
Shift 16 : uryyb
Shift 17 : tqxxa
Shift 18 : spwwz
Shift 19 : rovvy
Shift 20 : qnuux
Shift 21 : pmttw
Shift 22 : olssv
Shift 23 : nkrru
Shift 24 : mjqqt
Shift 25 : lipps
```

RESULT:

Thus, the additive cipher was successfully attacked by testing all 26 possible shifts.